

DETERMINISTIC PATH ENSEMBLES, DRIVEN COLLISION MAPS, AND ENDOGENOUS CONJUGATE PARAMETERS

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ABSTRACT. We replace every finite seam completion by a complete symbolic germ surface carrying all iterated forward and backward cuts. A depth-weighted current scale gives a spectral gap and finite-order response on the full dynamic singularity ledger. A symplectic router separates branch motions in an auxiliary canonical coordinate and yields an exact autonomous Hamiltonian suspension with full-port work.

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1. ROUND-EIGHT RECONSTRUCTION: COMPLETE DYNAMIC GERMS AND ROUTED HAMILTONIAN SUSPENSION

The finite biseam complex used in the preceding revision did not remember the oriented side of the branch preimages of the next vertical seam. We replace it by the complete symbolic germ surface. This simultaneously records every forward and backward singularity cut and turns the return map into an honest bijection before any transfer or response argument is made.

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1.1. The complete germ surface. Let $p_i(a) > 0$, $\sum_{i=1}^d p_i(a) = 1$, and $s_i(a) = \sum_{j \leq i} p_j(a)$. Put $\mathcal{A} = \{1, \dots, d\}$ and $\Omega = \mathcal{A}^{\mathbb{Z}}$ with shift σ . For $\omega \in \Omega$ define

$$q_a(\omega) = \sum_{n \geq 0} p_{\omega_0}(a) \cdots p_{\omega_{n-1}}(a) s_{\omega_n-1}(a),$$

$$p_a(\omega) = \sum_{n \geq 1} p_{\omega_{-1}}(a) \cdots p_{\omega_{-n+1}}(a) s_{\omega_{-n}-1}(a).$$

Empty products are one. At an endpoint the two symbolic expansions are not identified. The *complete dynamic germ surface* is the graph

$$\widehat{\Sigma}_a^\infty = \{(q_a(\omega), p_a(\omega), \omega) : \omega \in \Omega\}$$

with the topology transported from Ω . Its physical projection is $\pi_a(q, p, \omega) = (q, p)$.

Lemma 1.1 (All dynamic seams are present). *For every $N \geq 0$, the partition of $\widehat{\Sigma}_a^\infty$ into cylinders $[\omega_{-N}, \dots, \omega_N]$ refines*

$$\bigvee_{|j| \leq N} \pi_a^{-1} F_a^{-j} \mathcal{P}_a,$$

where $\mathcal{P}_a$ is the physical branch partition. In particular the one-step cuts $q = s_{i-1} + p_i s_j$, all of their iterated preimages, and all oriented sides at their intersections are distinct strata of the cylinder system.

Proof. The coordinate ω_n is exactly the branch of $F_a^n \pi_a(\omega)$. Therefore a cylinder fixes the side of each seam encountered for $-N \leq n \leq N$. Conversely, away from endpoints those branch labels determine the cylinder. At endpoints the symbolic expansion is retained rather than identified, so both oriented sides remain distinct. This proves the claimed refinement at every depth. $\square$

Define

$$\widehat{F}_a(q_a(\omega), p_a(\omega), \omega) = (q_a(\sigma\omega), p_a(\sigma\omega), \sigma\omega).$$

Theorem 1.2 (Complete-germ baker bijection). *The map $\widehat{F}_a$ is a homeomorphism of $\widehat{\Sigma}_a^\infty$, with inverse induced by σ^{-1} . Its physical projection is the generalized baker formula*

$$q' = \frac{q - s_{i-1}}{p_i}, \quad p' = s_{i-1} + p_i p, \quad i = \omega_0,$$

on every symbolic germ. Thus the formula is single valued on every preimage seam and every multiple seam crossing.

Proof. The series defining q_a, p_a converge uniformly because $\max_i p_i \leq 1 - (d-1)p_*$ on compact parameter sets. Substitution after a shift gives the displayed affine formula. The old branch is ω_0 and the new branch is ω_1 , including both endpoint expansions. Shift and inverse shift are continuous in the product topology, proving the assertion. $\square$

1.2. A depth-weighted current scale. For a finite word $w = \omega_{-r} \dots \omega_s$, let $[w]$ be its cylinder and $\partial^\alpha[w]$ the oriented parameter derivative current of order $|\alpha|$ obtained by differentiating the moving coding map π_a on that cylinder. Put $\ell(w) = r + s + 1$. Fix numbers

$$0 < \vartheta < \vartheta_+ < 1, \quad d\vartheta_+ < 1,$$

and define

$$\|u\|_{m,+} = \sum_{|\alpha| \leq m} \sum_w \vartheta_+^{\ell(w)} |\langle u, \partial^\alpha[w] \rangle|, \quad \|u\|_{m,-} = \sum_{|\alpha| \leq m-1} \sum_w \vartheta^{\ell(w)} |\langle u, \partial^\alpha[w] \rangle|.$$

The cylinder relations are quotiented before completion, so the same current is not counted under two refinements. Denote the completions by $\mathcal{B}_{a,m}^+$ and $\mathcal{B}_{a,m}^-$. The transfer operator is

$$\mathcal{L}_a u = u \circ \hat{F}_a^{-1}$$

with Bernoulli Jacobian weights when acting on densities.

Lemma 1.3 (Dynamic-cut Lasota–Yorke estimate). *For every finite m there are $C < \infty$ and $0 < \rho < 1$, uniform on a compact parameter interval, such that*

$$\|\mathcal{L}_a^n u\|_{m,+} \leq C\rho^n \|u\|_{m,+} + C\|u\|_{m,-}.$$

The inclusion $\mathcal{B}_{a,m}^+ \hookrightarrow \mathcal{B}_{a,m}^-$ is compact.

Proof. Shift removes one future symbol and adds one past symbol. After quotienting cylinder refinements, every depth- k coefficient has at most d children. The choice $d\vartheta_+ < 1$ absorbs this complexity. A parameter derivative creates only the two endpoint currents of the affected cylinder and lowers the available derivative order by one; these terms are placed in the weak norm. Iteration gives the displayed inequality. For compactness, truncate the cylinder depth at N and the derivative order at $m - 1$. The truncated range is finite dimensional, while the tail is bounded by $C(\vartheta/\vartheta_+)^N$. Hence the inclusion is a norm limit of finite rank maps. $\square$

Theorem 1.4 (Spectrum and finite-order response on all generated cuts). *The eigenvalue one of $\mathcal{L}_a$ is simple and the remainder of the spectrum on $\mathcal{B}_{a,m}^+$ lies in a disk of radius less than one. If G_a has m parameter derivatives in the depth-weighted current scale, then $a \mapsto \nu_a(G_a)$ is C^m . Every derivative is an absolutely convergent sum of cylinder interior terms and oriented currents on all iterated seams.*

Proof. Write a function as the sum of its martingale differences with respect to the future cylinders. On the mean-zero subspace, the shift removes the first martingale difference and the depth weight gives a strict contraction. A unit-modulus eigenvector has all martingale differences zero in both time directions and is constant. The compact embedding in Lemma 1.3 then gives a spectral gap.

In moving symbolic coordinates only the Bernoulli weights and the coding map depend on a . Differentiating a depth- N cylinder expectation gives

finitely many interior derivatives and endpoint currents. The number of depth- N terms grows at most like d^N , whereas their norm is bounded by ϑ_+^N and the resolvent contribution by ρ^N . The chosen weights make the differentiated series normally convergent through order m . $\square$

1.3. A symplectic router and an autonomous realization. The previous attempt incorrectly required branch motions with overlapping source and target regions to have disjoint support in one physical plane. We separate them in an auxiliary canonical routing coordinate. Let (y, η) be canonical coordinates and choose disjoint routing tubes $\mathcal{T}_i$ centered at $y = i$. A compactly supported Hamiltonian lift first moves the i th incoming strip into $\mathcal{T}_i$, a second Hamiltonian applies the affine exact symplectic branch map there, and a third lowers it to the i th outgoing horizontal port. The three stages occupy disjoint clock intervals, while the branch tubes are disjoint in (y, η) even when their physical projections overlap.

Lemma 1.5 (Exact symplectic routing). *There is a smooth periodic Hamiltonian $K_{a,\tau}(q, p, y, \eta)$ on a four-dimensional exact symplectic routing manifold such that its time-one map on the incoming section is the complete-germ baker map and its routing coordinate returns to $(y, \eta) = (0, 0)$. The construction is C^{m+1} in a and has matching flat jets on every symbolic germ.*

Proof. The lift and lowering stages are translations generated by cutoff multiples of η . Inside tube i , the affine branch has generating function

$$S_i(q, P) = p_i^{-1}(q - s_{i-1})(P - s_{i-1}),$$

which generates $Q = (q - s_{i-1})/p_i$ and $p = (P - s_{i-1})/p_i$. Interpolating S_i to the identity with a clock cutoff gives an exact symplectic isotopy. All branch isotopies are disjoint in the routing coordinate. Flat cutoffs make their jets agree at tube boundaries. The final lowering is the inverse of the lift, so the auxiliary coordinates close exactly. $\square$

On the extended phase space with clock pair (τ, E) set

$$H_a = E + K_{a,\tau}.$$

Theorem 1.6 (Autonomous routed suspension). *The Hamiltonian flow of H_a has exact first return time one to $\{\tau = 0, y = \eta = 0\}$ and return map $\hat{F}_a$. It preserves Liouville flux and realizes every dynamic germ, not merely a positive-measure core.*

Proof. Hamilton's equation gives $\dot{\tau} = 1$. During one clock period the lift, branch generating-function isotopy, and lowering occur in that order. The router lemma identifies their composition with $\hat{F}_a$ and returns the auxiliary pair to zero. Each stage is exact symplectic, hence the section flux is preserved. Endpoint expansions occupy distinct routed germs and therefore never require an ambiguous choice. $\square$

1.4. Work, calibration, and universality. Place a normalized work form $\epsilon_i \rho_i(\tau) d\tau$ in the incoming routing tube before the branch motion.

Proposition 1.7 (Full-port work). *For every return segment with incoming symbol i ,*

$$\int_0^1 \epsilon_i \rho_i(\tau) \dot{\tau} dt = \epsilon_i.$$

Thus the work is the additive cylinder cocycle on the complete germ surface, including every seam germ.

Proof. The symbol is constant during the work stage and $\dot{\tau} = 1$. The integral of ρ_i is one. Concatenation proves the cocycle identity. $\square$

Theorem 1.8 (Finite-alphabet universality and conditional calibration). *Every smoothly parameterized finite Bernoulli shift with positive weights and every finite additive port cocycle admits the complete-germ routed Hamiltonian realization above. A cash-additive law-invariant time-consistent certainty equivalent has the same coefficient as the mechanical exponential likelihood exactly when equality of the two normalized likelihood cocycles is imposed.*

Proof. The coding, depth-weighted transfer scale, and router construction used only finiteness and positivity of the alphabet weights. For calibration, cash additivity and time consistency give constant absolute risk aversion. Equality of normalized likelihoods on a nonconstant port cocycle forces equality of the two exponential coefficients; without that compatibility axiom no such identification follows. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared local source, short-time, finite-horizon or Boltzmann–Grad regime specified in its theorems. Every load-bearing cross-paper input is typed in the round-eight dependency ledger. Clean compilation and automated structural checks are necessary verification gates; independent external journal review remains a separate process.

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